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ABSTRACT

As one of the leading technologies for chemical enhanced oil recovery (cEOR), surfactant-polymer (SP) flooding technology has long attracted the interest of petroleum scientists and engineers. However, most of its application scenarios are based on the five-spot well pattern. The EOR potential in an inverted seven-spot well pattern is seldom ever recorded. The applicability of the SP system in the inverted seven-spot well pattern was examined based on the physical characteristics of Karamay Oilfield in China. The numerical simulation and the one-dimensional core flooding experiment were used to compare the sweep intensities and EOR abilities of the two well patterns. The migration law and the EOR ability of the SP system were assessed by a specially made one-third inverted seven-spot configuration. The main controlling factors and compatibility charts of SP flooding development in the inverted seven-spot well pattern were obtained. Results show that 61% of the region is represented by a weak swept state in the inverted seven-spot well pattern. The effective swept area is greatly increased by appropriately raising the viscosity and slug size of the SP system. Compared to constant viscosity injection, step-down viscosity injection further increases the sweep range and oil recovery. The inverted seven-spot well pattern has a greater swept area of the SP system than the five-spot one, but a weaker strength. Polymer concentration is the most effective factor of SP flooding in the inverted seven-spot well pattern, followed by oil viscosity and surfactant concentration. The study can broaden the application of the SP system in the inverted seven-spot well pattern.

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I. INTRODUCTION

Due to exponential population growth and rapid industrialization, crude oil is deemed the primary energy source for humankind. It will continue to rule the energy market in the near future until more affordable and viable options for energy generation are found. Compared with other mineral resources, oil recovery is relatively low. The oil production obtained by natural energy (primary oil recovery) and water injection (secondary oil recovery) is limited.¹ In such cases, a sizeable amount (about 65%–77%) of the remaining oil is still confined in reservoirs.^{2–6} As a tertiary oil recovery technology, the SP (surfactant-polymer) flooding technology gives full play to the synergistic effect of surfactant reducing oil–water interfacial tension and polymer improving sweep volume to enhance oil recovery (EOR).^{7–13} By adjusting the oil–water mobility ratio and reducing the permeability of the water phase, the SP solution can effectively increase the sweep volume.^{14,15}

Additionally, altering the rock wettability and decreasing the residual oil absorption capacity can enhance the oil displacement efficiency in porous media.^{8,16} So that the EOR purpose is fulfilled. In the 1960s, a series of SP flooding pilot tests were carried out by the United States and have achieved remarkable results.^{17,18} Subsequently, in the 1980s, development test areas were successively deployed in Shengli, Xinjiang, and Daqing Oilfields in China, all of which have realized positive development results and economic benefits.^{19–21} These results provide strong theoretical and technical justification for reducing water cut and increasing oil recovery in mature water-injected oilfields.¹⁰ The additional oil recovery of SP flooding is expected to be 13%–18% of the original oil in place (OOIP), between those of polymer flooding and ASP (alkali-surfactant-polymer) flooding.²² Five components make up the primary technical characteristics of SP flooding: (1) excellent EOR potential and a wide range of applicable reservoir conditions;

(2) the avoidance of alkali issues compared to ASP flooding;²³ (3) the ease of the ground injection process, which simplifies the injection process and anti-corrosion treatment;²² (4) good economics, which lowers the costs of chemical agents, equipment, and operation;²² and (5) higher requirements for the surfactant, which makes it more challenging to achieve ultra-low interfacial tension under alkali-free conditions.²²

The SP flooding technology has made a breakthrough in the field test of conglomerate reservoirs in Xinjiang Karamay Oilfield of China.^{8,24} It has been identified as the main technology for greatly enhancing oil recovery in Xinjiang conglomerate reservoirs. Since 2015, large-scale promotion and development of SP flooding have been done in the Badaowan Formation of Well Block 530 and the Kexia Formation of Qixi Block, with impressive results.^{8,25} However, as the depth of the reservoir increases, the productivity footage ratio decreases and the economic gain decreases. Dynamic adjustment of the existing well pattern is required to achieve effective development of the conglomerate reservoir after water flooding and stimulation of the production-declined reservoir.^{26–38} Compared with the conventional five-spot well pattern, the inverted seven-spot well pattern is a type of triangular well pattern system, where the spacing between any adjacent well points is equal, making the well pattern arrangement more regular. In addition, it allows for the placement of additional wells in a given region, substantially increasing the well spacing density and enhancing reservoir development effectiveness. It will, however, significantly increase production costs and design complexity. Moreover, the inverted seven-spot well pattern has an injection-production well ratio of 1:2 (for the five-spot one, the ratio is 1:1).³⁹ Therefore, the injection fluid has a larger swept area but a weaker intensity. The application potential of the SP system will inexorably be affected by this inherent feature. It is crucial to assess the adaptability and the development benefit of the SP system in an inverted seven-spot well pattern.⁴⁰

The Badaowan Formation is composed of five oil layers, J_1b_1 , J_1b_2 , J_1b_3 , J_1b_4 , and J_1b_5 , with an average permeability of $173.91 \times 10^{-3} \mu\text{m}^2$.⁸ J_1b_4 and J_1b_5 (often abbreviated as J_1b_{4+5}) are the target horizons for chemical flooding. The upper sandstone section and the lower conglomerate section make up the two lithological divisions. In this research, various numerical and laboratory methods were adapted based on the reservoir's physical characteristics of J_1b_{4+5} . First, the pressure and pressure gradient distributions of two well patterns were analyzed by numerical simulation, and the various sweep intensity regions were separated. Second, the one-dimensional core displacement experiment was conducted to determine the SP system's capacity to recover crude oil at various sweep intensities by regulating the displacement rate. On that basis, the EOR potential in the two well patterns was compared. Third, the migration law and the oil displacement efficiency of the SP system in the inverted seven-spot well pattern were clarified by the custom-made flat core displacement experiments. After selecting the appropriate injection method, the EOR capabilities of the two well patterns were compared in detail.

Finally, the oil recovery fitting formula was developed by multiple linear regression. The main controlling factors of SP system development in the inverted seven-spot well pattern were obtained by Pearson correlation coefficient.

II. EXPERIMENTAL

A. Materials

Hydrophobic associated polymer KYPAM (kangyan polyacrylamide, a kind of anti-salt polyacrylamide) with an average molecular weight of $1.62 \times 10^7 \text{ g mol}^{-1}$ and a hydrolysis degree of 26.76% was supplied by Beijing Hengju Chemical Group Co., Ltd. Naphthenic petroleum sulfonate KPS (Karamay petroleum sulfonate) with a molecular weight of 452 g mol^{-1} and an active content of 30.05 wt. % was supplied by Xinjiang Jinta Investment (Group) Co., Ltd.

The dehydrated crude oil (density of 0.8763 g cm^{-3}) was obtained from Karamay Oilfield, China, with a viscosity of 39.54 mPa s measured by concentric cylinder system CC27 of Physica MCR301 (Anton Paar, Austria) at the reservoir temperature (35.6°C) and the shear rate of 7.34 s^{-1} . The synthesized formation water with total dissolved solids (TDS) of $14\,466.24 \text{ mg L}^{-1}$ was prepared according to the ion composition of natural formation water in Karamay Oilfield, as indicated in Table I.

B. Comparison of sweep intensity under different well pattern conditions

The inverted seven-spot and five-spot well patterns were simulated using tNavigator under the condition of single-phase water flow, and MATLAB was used for post-processing to clarify the impact of well pattern type on the sweep intensity and obtain its distribution law. The producer–injector spacing was adjusted to 150 m for comparison. The injection well was injected quantitatively, while the production wells maintained constant pressure. Figures 1(a) and 1(b) show the obtained pressure distributions. Once the pressure data had been gathered, the pressure gradient distributions were calculated by the central difference method, which were shown in Figs. 1(c) and 1(d).

C. EOR efficiency of SP system in the one-dimensional core flooding

One-dimensional core (artificial sandstone cores, diameter $d = 2.5 \text{ cm}$, length $L = 10 \text{ cm}$) displacement tests at 35.6°C were used to evaluate the EOR capabilities of the SP system under various sweep intensities. Figure 2(a) depicts the experimental apparatuses for this section. First, the brine mentioned in Table I was used to dissolve 0.3 wt. % KPS, and then, 1500 and 2200 mg L^{-1} KYPAM were added to prepare two SP systems with different viscosities (the low viscosity with 20.2 mPa s and the high viscosity with 39.8 mPa s at the reservoir temperature of 35.6°C and the shear rate of 7.34 s^{-1} , respectively). The core was then vacuumed and soaked with brine and oil in sequence, followed by two days of aging. Finally, the core was flooded

TABLE I. Karamay Oilfield synthesized water.

Composition	$\text{K}^+ + \text{Na}^+$	Mg^{2+}	Ca^{2+}	Cl^-	SO_4^{2-}	HCO_3^-	TDS	pH
Concentration (mg L^{-1})	5145.66	27.04	208.06	7388.7	22.26	1674.52	14 466.24	7.45

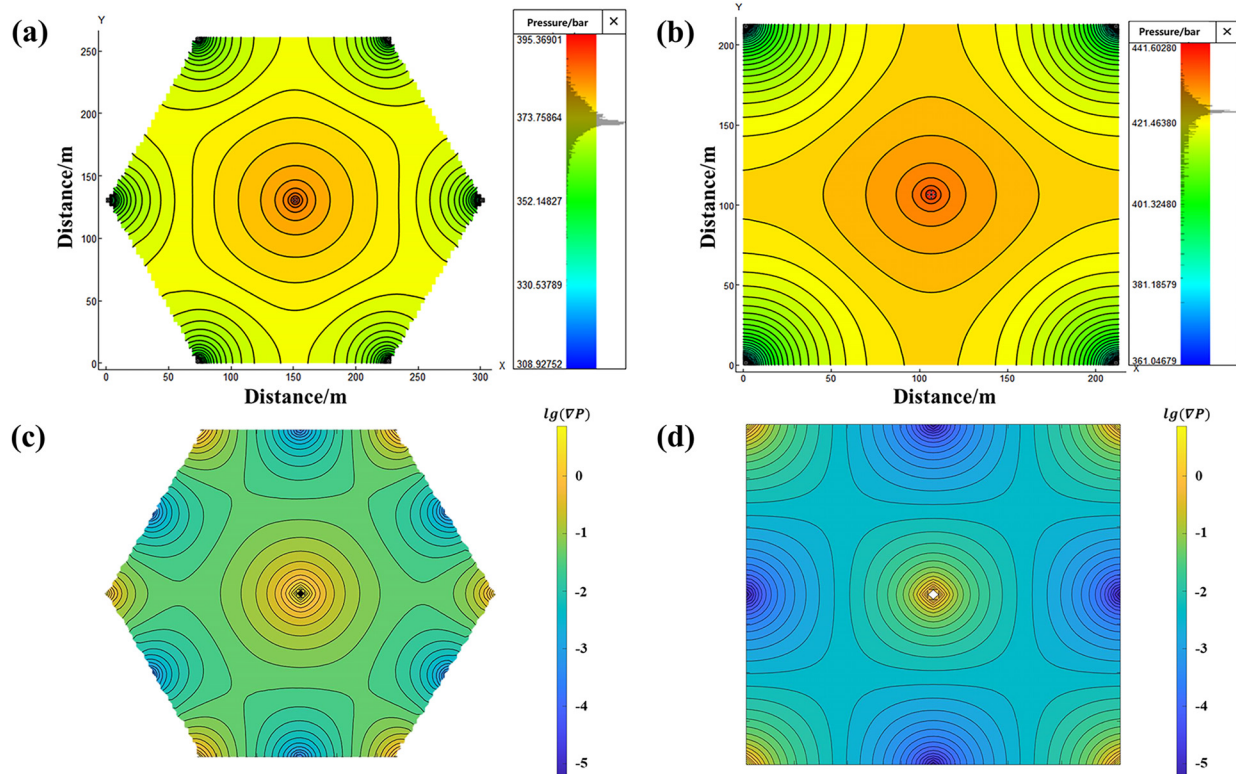


FIG. 1. Pressure distributions of (a) inverted seven-spot and (b) five-spot well patterns; pressure gradient distributions of (c) inverted seven-spot and (d) five-spot well patterns.

by the SP system solution at varied flow rates until the water cut reached a value higher than 98%. The injection pressure and oil recovery were tracked throughout the experiment. The variables, which are presented in Table II for a total of twelve experimental sets, are injection rate, permeability, and displacement phase viscosity.

D. Migration characteristic and EOR potential of SP system in the inverted seven-spot well pattern

1. Design and fabrication of flat cores

A one-third inverted seven-spot configuration, sketched in Fig. 3(a), was designed to investigate the migration law of injection fluid in the inverted seven-spot well pattern and the oil displacement efficiency under various injection strategies. In accordance with the streamline distribution and injector–producer ratio in the well pattern, the designed core (the corresponding regular hexagon had a side length of 24 cm) was cut from a flat core (length 30 cm $\times$ width 30 cm $\times$ thickness 4.5 cm, permeability $300 \times 10^{-3} \mu\text{m}^2$, porosity 24.1%). An injection well, two production wells, and several electrodes were positioned on the core according to the positions shown in Fig. 3(a). The electrodes were used to monitor the change of electrical resistance in each position in real-time. An inversion calculation from the resistivity-saturation standard curve was adopted to depict the oil saturation distribution at different stages.

To further compare the differences in the migration law and the EOR capability of injection fluid in different well pattern types, a one-fourth five-spot configuration (permeability $300 \times 10^{-3} \mu\text{m}^2$, porosity

24.1%) with one injection well, one production well, and several electrodes was designed and fabricated in accordance with the same concept, as shown in Fig. 3(b). According to the principle of equal control reserves for injection wells in two corresponding complete well patterns, width, and thickness of the one-fourth five-spot configuration were 19.5, 19.5, and 4.5 cm, respectively.

2. Oil displacement experiments in the flat cores

This section sought to investigate the SP system's migration traits and EOR potential in the inverted seven-spot well pattern. Figure 2(b) illustrates the flat core flooding and data acquisition systems. The injection rate was set at 0.3 ml min^{-1} , and the temperature was maintained at 35.6°C . The brine, SP system solution (fixed 0.3 wt. % KPS and varying concentrations of KYPAM), and crude oil were utilized here. The SP system slug design was stated in Table III. Experiments (1) and (2) were used to compare the effect of the viscosity of SP system on its development effect, while experiments (2) and (3) were used to compare the effect of injection volume. Moreover, experiments (2), (4), and (5) were used to optimize the injection method of the SP system. For comparability of the results, the total amount of polymer used for SP slugs in experiments (2), (4), and (5) was almost equal. The viscosities of the three SP system slugs from low to high were: 20.2, 39.8, and 54.7 mPa s. The cores were vacuumed and saturated with the synthesized water to establish the initial conditions. Once brine production ceased, the core was filled with filtered crude oil and matured for two days. The initial oil saturation S_{oi} of the flat core was

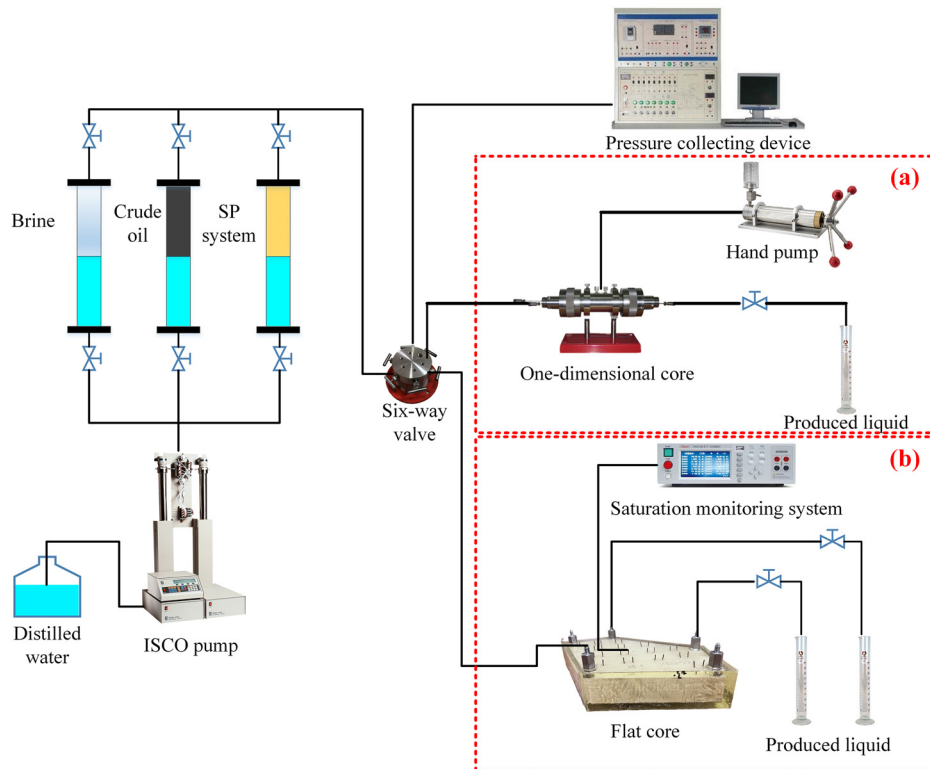


FIG. 2. Diagram of the oil displacement experiments in (a) the one-dimensional core and (b) the flat core.

TABLE II. Schemes of the EOR capacity experiments by one-dimensional core flooding.

No.	Permeability/ $\times 10^{-3} \mu\text{m}^2$	Porosity/ %	Oil saturation/ %	Viscosity/ mPa s	Injection rate/mL min^{-1}
1	50.3	24.17	70.34	20.2	0.05
2	49.3	24.04	70.58	20.2	0.1
3	48.8	23.79	71.39	20.2	0.3
4	48.7	23.27	69.71	39.8	0.05
5	50.1	24.19	70.17	39.8	0.1
6	50.7	24.57	69.87	39.8	0.3
7	300.0	23.49	72.97	20.2	0.05
8	299.8	24.02	73.14	20.2	0.1
9	298.7	24.67	74.07	20.2	0.3
10	297.1	23.18	73.19	39.8	0.05
11	302.3	24.45	72.57	39.8	0.1
12	301.5	24.37	73.56	39.8	0.3

around 70%. The specific experimental procedure was as follows: (1) inject the synthesized water until maximum water flooding recovery was attained; (2) inject a certain pore volume (PV) of SP system solution; and (3) perform the subsequent water flooding to accomplish the ultimate oil recovery.

During the experiments, liquid production, injection pressure, and electrical resistance were measured. The ideal SP system injection

strategy was obtained by contrasting the oil recovery values. A similar methodology was adopted in the five-spot well pattern core to compare the SP system's migration law and oil displacement efficiency in different well patterns using the obtained optimal strategy.

E. Establishment of SP system compatibility chart in the inverted seven-spot well pattern

The numerical simulation of SP flooding in an inverted seven-spot well pattern model (one injector and six producers) in an infinite formation was investigated by CMG-STARS simulator. Based on previous experience and literature research, five factors (three fluid parameters: oil viscosity, KYPAM concentration, and KPS concentration; two injection parameters: injection timing and volume) were selected for research in this part. As shown in Table IV, the orthogonal scheme with five factors and four levels was designed to investigate the adaptability of SP system in the inverted seven-spot well pattern. In order to obtain the specific time of different injection timing and compare with traditional water flooding, it is necessary to supplement the numerical simulation of four groups of crude oil with different viscosity under traditional water flooding development mode. A total of 20 groups of experiments were therefore conducted.

The model was designed according to the geology data of J_1b_{4+5} and the fluid property parameters obtained using laboratory testing methods. As shown in Fig. 4, the total number of grids was 51 510, the number of active grids was 38 510, and the injector-producer spacing was 300 m. The model's thickness was 30 m, divided into ten layers from top to bottom, and the thickness of each layer was 3 m. The

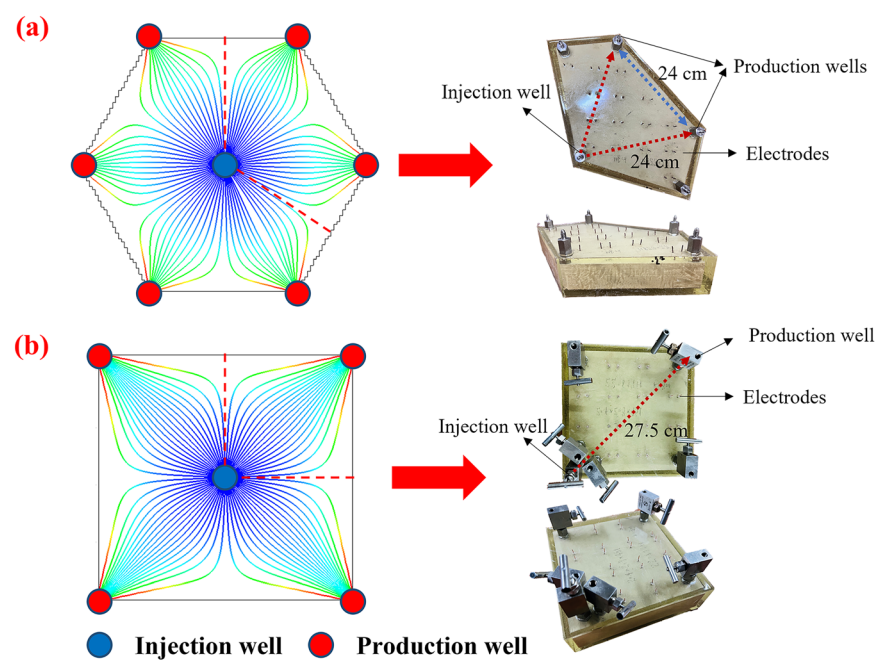


FIG. 3. Design of (a) inverted seven-spot and (b) five-spot well pattern cores.

TABLE III. Schemes of the EOR capacity experiments under various injection methods based on the inverted seven-spot well pattern.

No.	Injection strategy	SP system slug design	
		KPS concentration	KYPAM concentration, viscosity, and slug size
1	Constant viscosity injection	0.3 wt. %	1500 mg L ⁻¹ (20.2 mPa s, 0.9 PV)
2	Constant viscosity injection	0.3 wt. %	2200 mg L ⁻¹ (39.8 mPa s, 0.9 PV)
3	Constant viscosity injection	0.3 wt. %	2200 mg L ⁻¹ (39.8 mPa s, 0.7 PV)
4	Step-up viscosity injection	0.3 wt. %	1500 mg L ⁻¹ (20.2 mPa s, 0.2 PV) + 2200 mg L ⁻¹ (39.8 mPa s, 0.5 PV) + 2700 mg L ⁻¹ (54.7 mPa s, 0.2 PV)
5	Step-down viscosity injection	0.3 wt. %	2700 mg L ⁻¹ (54.7 mPa s, 0.2 PV) + 2200 mg L ⁻¹ (39.8 mPa s, 0.5 PV) + 1500 mg L ⁻¹ (20.2 mPa s, 0.2 PV)

TABLE IV. Five factors and four levels variable design table.

Factor	Value/range	Variables
Oil viscosity/mPa s	7.6, 48, 100, 200	4
KYPAM concentration/wt. %	0.1, 0.15, 0.22, 0.3	4
KPS concentration/wt. %	0.1, 0.2, 0.3, 0.5	4
Injection timing (water cut)/%	85, 90, 95, 98	4
Injection volume/PV	0.8, 0.9, 1.0, 1.2	4
Total	/	16 + 4

upper five layers were set as sandstone layers with the permeability of $300 \times 10^{-3} \mu\text{m}^2$, porosity of 0.24, and oil saturation of 70%. The lower five layers were set as conglomerate layers, with the permeability of $50 \times 10^{-3} \mu\text{m}^2$, porosity of 0.22, and oil saturation of 60%. The injector was arranged in the center of the model, and the six producers were set at six corners of the hexagon. All wells are set to shoot all layers.

The injector’s injection rate and the producers’ recovery rate were set to 300 and 50 m³/d, respectively. The development cycle was set to 50 years.

III. RESULTS AND DISCUSSION

A. Sweep intensity division under different well patterns

As shown in Fig. 1(c), seven high-pressure and six low-pressure gradient zones are in the inverted seven-spot well pattern. Injection and production wells serve as the centers of high-pressure gradient zones, while equilibrium points on the connecting line between nearby oil wells serve as the centers of low-pressure gradient zones. Similarly, the five-spot one features four low-pressure gradient zones and five high-pressure gradient zones [Fig. 1(d)], respectively. Referring to the oilfield development data, the sweep intensity of different regions in the well pattern is divided according to the pressure gradient. The specific division basis is detailed in Table V.

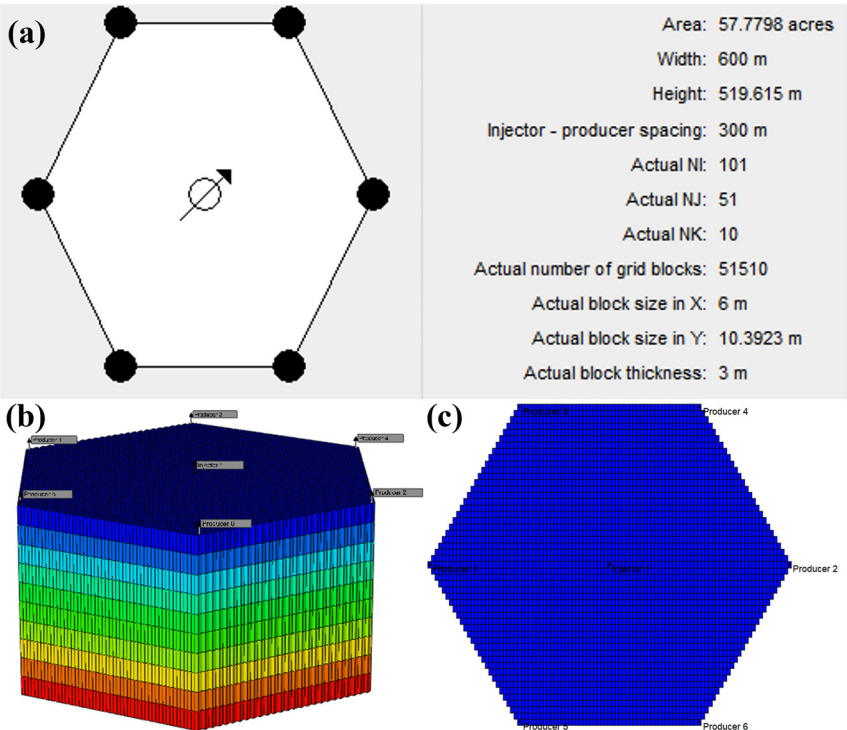


FIG. 4. Inverted seven-spot well pattern numerical model: (a) basic parameters, (b) main view, and (c) top view of the model.

TABLE V. Classification standard of different sweep intensity regions.

Classification	Strong swept region	Medium swept region	Weak swept region	Ineffective swept region
Criterion	>0.2 MPa/m	0.1–0.2 MPa/m	0.03–0.1 MPa/m	<0.03 MPa/m

It can be seen from Figs. 5(a) and 5(b) that the strong swept regions are almost distributed in the bottom hole and surroundings; the medium swept regions of the seven-spot well pattern are distributed in the near-well zone, while those of the five-spot one are mainly distributed on the producer–injector connection line; the weak swept regions of the seven-spot well pattern are distributed in the far-well

zone, while those of the five-spot one are near the connecting line of the production wells; in addition, the ineffective swept regions are all distributed on the connecting line of the production wells. By comparing the bar graph of the sweep intensity classification [Fig. 5(c)], it can be found that the inverted seven-spot well pattern presents a weak swept state overall, with the weak swept region accounting for 61% of

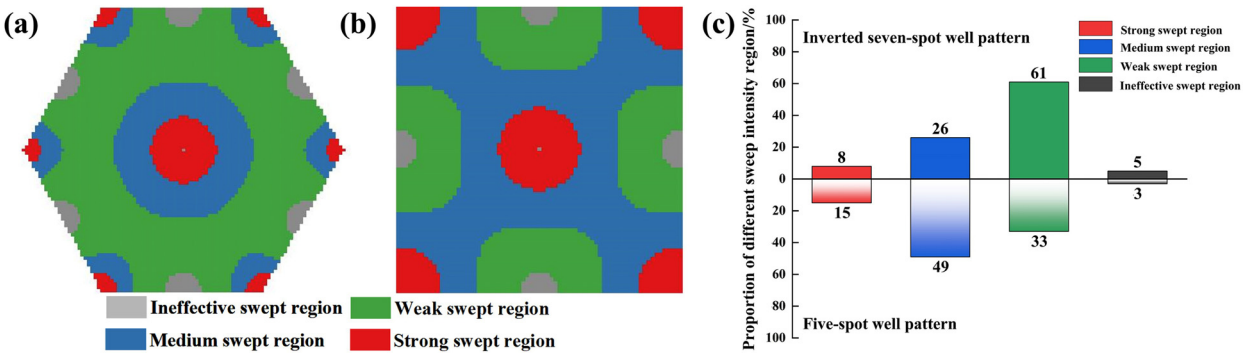


FIG. 5. Pressure gradient distributions of (a) inverted seven-spot and (b) five-spot well patterns, and (c) difference of two well patterns in different sweep intensity region proportion.

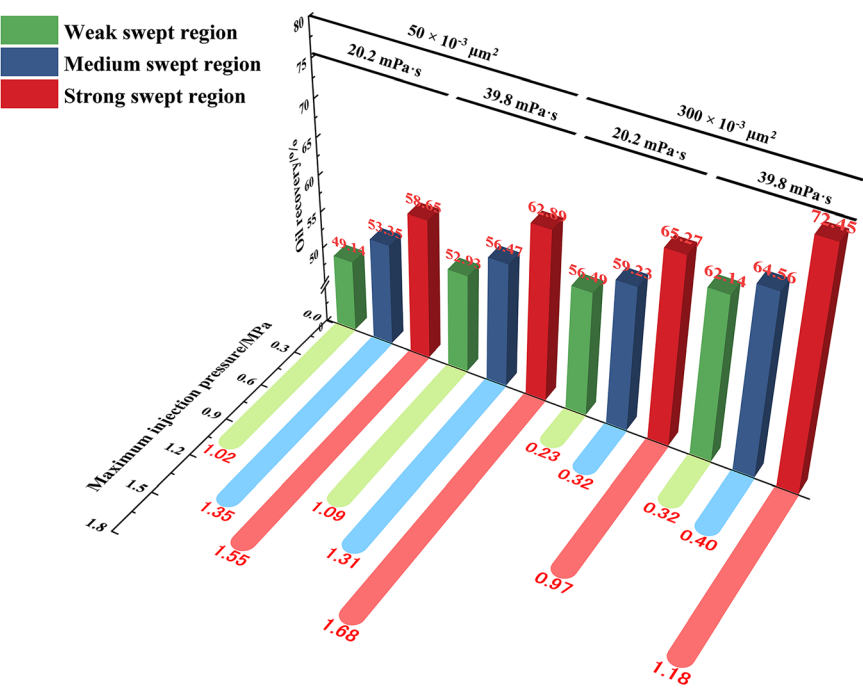


FIG. 6. Difference in EOR capacities of two SP systems under different sweep intensities.

the total, which is almost twice that for the five-spot one. The difference in sweep intensity is bound to be reflected in the EOR capacity of the SP system, necessitating a study into the system’s adaptation to different well patterns.

B. Influence of sweep intensity on EOR efficiency

In this section, the crude oil recovery capacity of the SP system under varied sweep intensities was obtained by one-dimensional core experiments, and the difference in EOR potential in different well patterns was given in combination with the results of Sec. III A. Different injection rates were used to represent the sweep intensities in different regions of the well patterns (Fig. S1). The strong sweep intensity was indicated by 0.3 ml min^{-1} , the medium by 0.1 ml min^{-1} , and the weak by 0.05 ml min^{-1} . Figure 6 lists the EOR capacities of the two SP systems (20.2 and 39.8 mPa s, respectively) under different sweep intensities. In the same circumstances, the high-viscosity system had a higher EOR capacity than the low-viscosity one. Moreover, the EOR capacity was greatly improved with an increase in sweep intensity, and the improved range was 15%–20% regardless of whether the systems were high- or low-viscosity. Furthermore, by comparing the EOR ability and maximum injection pressure under two permeability conditions, it can be found that the SP system performed better under the high permeability condition, and the corresponding injection pressure was lower, indicating that the system was easier to penetrate the high-permeability layer.

Concerning the whole well pattern, the difference in oil recovery induced by varying sweep intensities will inevitably be reflected in the EOR potential of the SP system in different well patterns. The data in Fig. 6 are weighted and averaged in accordance with the different sweep intensity region proportions divided in Fig. 5(c) to obtain the EOR potential of the SP system in the whole well patterns. The results

are shown in Table VI. It is not difficult to find that due to the dominance of the weak swept region in the inverted seven-spot well pattern, the EOR potential of the SP system in it is lower than that in the five-spot one, and the difference is about 4% points.

C. Influence of viscosity and injection volume on SP flooding in inverted seven-spot well pattern

The sweep ability of SP systems with different viscosities in the inverted seven-spot well pattern was investigated. Figure 7 lists the oil saturation distribution before and after displacement by SP systems under 0.9 PV constant viscosity injection. The average oil saturation $\bar{S}_o$ after water flooding was about 36% [36.1% in Fig. 7(a) and 35.9% in Fig. 7(b)]. Furthermore, the oil saturation on the connecting line of injection–production wells was comparatively low, suggesting that a certain dominating channel had evolved after water flooding. As shown in Fig. 7(a), the low-viscosity system could inhibit water

TABLE VI. Calculating EOR potential of SP systems in two well patterns.

Permeability/ $\times 10^{-3}\text{ }\mu\text{m}^2$	SP system viscosity/mPa s	Calculating EOR potential/%		
		In the inverted seven-spot well pattern	In the five-spot well pattern	Difference
50	20.2	48.54	52.87	−4.33
50	39.8	52.00	56.36	−4.36
300	20.2	55.08	59.04	−3.96
300	39.8	60.49	64.86	−4.37

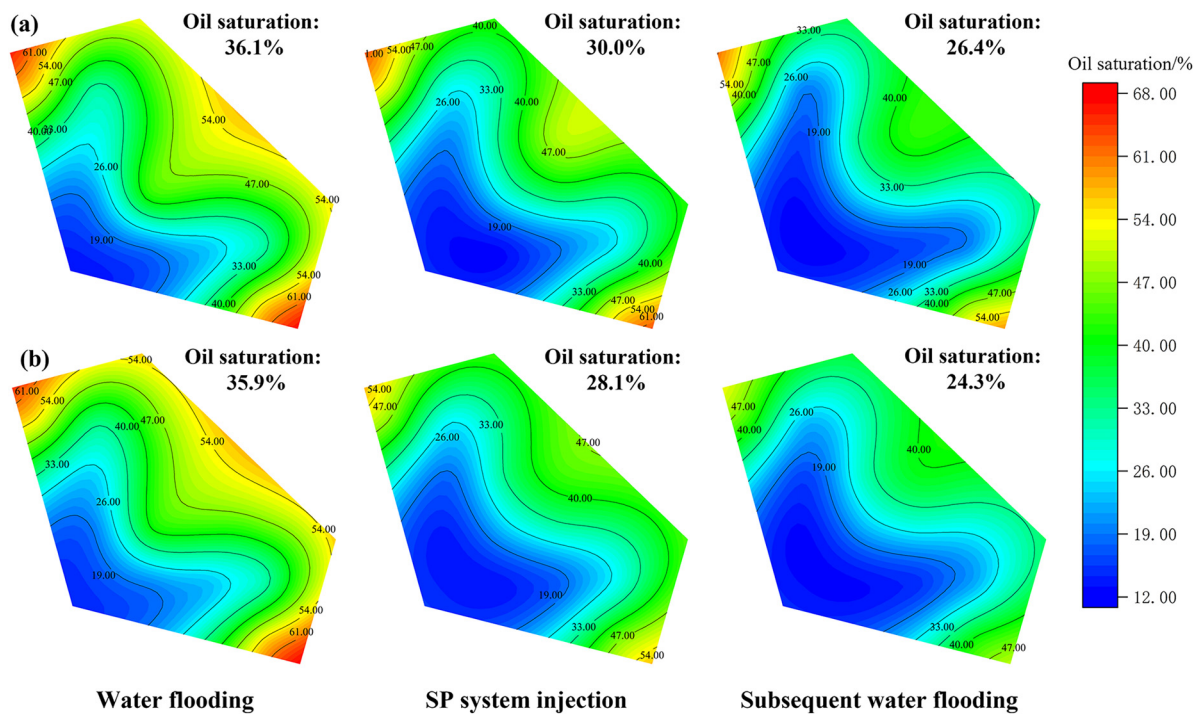


FIG. 7. Oil saturation distributions of inverted seven-spot well pattern before and after displacement by (a) low- (20.2 mPa s) and (b) high-viscosity (39.8 mPa s) systems under 0.9 PV constant viscosity injection.

channeling to a certain extent, so the effective swept area increased. The oil content remained high in the unswept area. The water broke through quickly when undertaking subsequent water flooding and was unable to further affect the non-primary streamline area. The $\bar{S}_o$ of the well pattern dropped by 9.7% points, to 26.4%. In comparison with the low-viscosity system, the high-viscosity one could successfully prevent water break through, increasing the effective swept area and reducing the oil saturation in the unswept area substantially [Fig. 7(b)]. In the subsequent water flooding stage, the water break-through was delayed, which further decreased the oil saturation in the non-primary streamline area. The final oil saturation $\bar{S}_o$ was 24.3%, which was 2.1% points less than that handled by the low-viscosity system. It indicates that the viscosity of the SP system applied to the inverted seven-spot well pattern should be appropriately increased under the premise of satisfying the formation compatibility.

In order to compare more intuitively, the oil saturation before and after the SP system's displacement was subtracted, and the resultant value was then divided by the pre-displacement saturation (the oil saturation after water flooding) to determine the percentage of oil saturation reduction in each area of the well pattern, as shown in Fig. 8. According to Figs. 8(a) and 8(b), compared to water flooding, SP flooding mostly affected the reservoir model's front and center sections as well as the major flow channel area. Compared to the low-viscosity system, the high-viscosity SP system could establish a thicker "oil bank" and significantly increase the effective swept area. Using a standard of oil saturation reduction of over 20%, the effective swept areas of the low- and high-viscosity SP systems accounted for 46.7% and

66.7%, respectively. From Fig. 8(c), it can be seen that the two different viscosity systems had similar displacement efficiency for the strong swept region; compared to the low-viscosity system, the high-viscosity one had a significantly enhanced displacement ability for the remaining oil in the medium/weak swept regions, with increases of 17.35% and 38.31% in the changes of oil saturation percentage, respectively.

Then, the influence of injection volume on the development effect of SP systems was clarified. Figure 9 depicts the variation in oil saturation in the reservoir model before and after the injection of a 0.7 PV high-viscosity SP system. By comparing Fig. 7(b) with Figs. 9(a)–9(c), it can be observed that during the water flooding stage, the water front mainly advanced along the mainstream direction, while the displacement efficiency of the crude oil in the non-mainstream area was relatively low, with a maximum oil saturation of greater than 61%. However, after injecting the SP fluid, the overall oil saturation of the well pattern was dramatically reduced, and the SP system reduced the water–oil mobility ratio, which to a certain extent hindered the channeling in the mainstream direction. The crude oil in the non-mainstream and unswept areas was also noticeably displaced. Compared with the 0.7 PV slug, the 0.9 PV slug had a more substantial impact, and the oil saturation could be further reduced. By referring to Figs. 8(b) and 9(d), it can be observed that the 0.9 PV SP system could further decrease the oil saturation in the same region, significantly increase the efficient displacement area, and increase the oil recovery in the hard-to-reach areas. Indicating that increasing the injection volume can aid in recovering the residual oil in the unswept area, and the effective swept areas of the 0.7 and 0.9 PV SP systems were 40% and 66.7%, respectively.

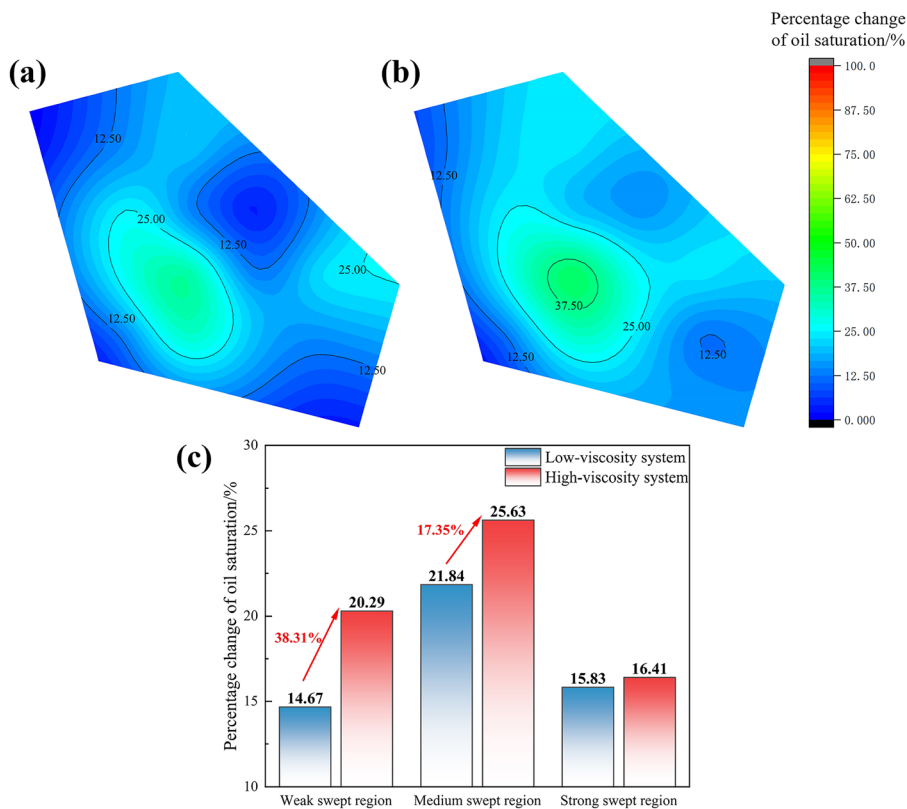


FIG. 8. Percentage change of oil saturation before and after (a) low- and (b) high-viscosity SP flooding, and (c) the comparison in different sweep intensity regions.

D. Optimization of injection methods and EOR ability in different well patterns

Evaluation was done on how injection methods affected the EOR capability of SP systems in the inverted seven-spot well pattern. Figures 10(a)–10(c) show the dynamic production curves of the inverted seven-spot well pattern model under three different SP injection strategies with the same injection volume (0.9 PV). The inverted seven-spot well pattern's water-free stage (prior to water break through) was found to be 0.133 PV under testing conditions, and the associated recovery rate was around 21%. The water cut achieved 95% and the water flooding recovery was about 44.50% after 0.933 PV of water flooding. Subsequently, various SP system injection strategies were applied to the model (as stated in Table III). The EOR values from high to low were: step-down viscosity injection (22.02%) > constant viscosity injection (18.05%) > step-up viscosity injection (15.90%). In conjunction with the water content variation curve, it was found that the decreased range of water cut and the appearance time of the lowest water cut were positively correlated with the EOR capacity. The phenomenon suggests that the high-viscosity slug injected preferentially can form a highly concentrated oil bank at the displacing front, which has a stronger mobility control ability than the low-viscosity one, reducing the water cut quickly and efficiently, and increasing the sweeping area of the injected fluid in the well pattern. Therefore, under the condition of a fixed slug size of the SP system, step-down viscosity injection should be preferred to enhance oil recovery further.

For further evaluation of the fluid migration law and the EOR ability of the SP system in different well patterns, the dynamic production curve and the oil saturation distribution under the optimal injection method (step-down viscosity injection) were listed in Figs. 10(c), 10(d), and 11. The five-spot well pattern's water-free stage was 0.1 PV, and the associated oil recovery was around 11%. After 0.8 PV of water flooding, water flooding recovery reached 41.45%. The water flooding stage was shorter and the recovery rate was lower compared to the inverted seven-spot well pattern. In addition, as depicted in Fig. 11, the water channel in the five-spot well pattern was more apparent, and the residual oil saturation was high outside of the channel area ($\bar{S}_o > 68\%$ in the unswept areas). After SP system injection and subsequent water flooding, the ultimate oil recovery of the five-spot well pattern was 66.09%, almost the same as that of the inverted seven-spot one. The enhanced oil recovery factor was 24.64% and 2.62% points higher than that in the inverted seven-spot one. Apparent differences can be found in the sweeping area and displacement efficiency of the SP system in the two well patterns (Fig. 12). From Fig. 12, the SP system in the inverted seven-spot well pattern had a wide range of spread, including the middle of the model, the primary streamline area, and the unswept area by water flooding. However, due to the small proportion of high-efficiency swept area (green area in the figure), the EOR capacity of the SP system in the inverted seven-spot well pattern was relatively low. Compared with the inverted seven-spot well pattern, the swept area and the production efficiency of the SP system in the five-spot one were more highly concentrated in the primary streamline

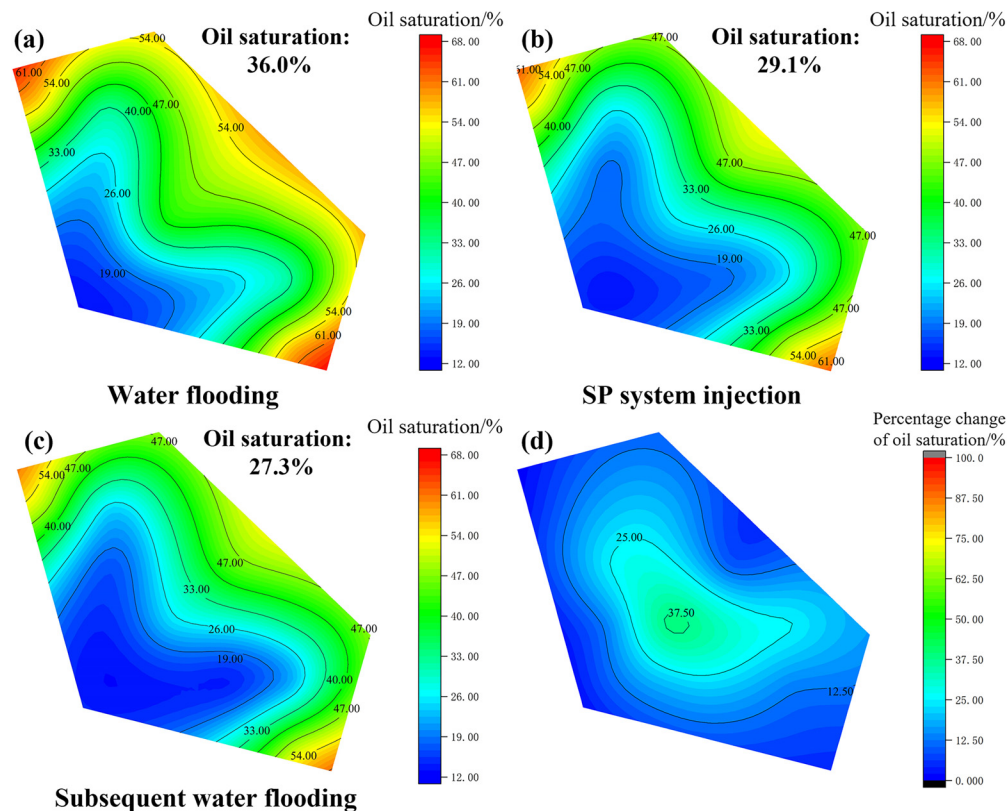


FIG. 9. Oil saturation distributions after (a) water flooding, (b) high-viscosity (39.8 mPa s) SP flooding, (c) subsequent water flooding, and (d) percentage change of oil saturation under 0.7 PV constant viscosity injection.

area, while the corner area was less utilized. The effective swept areas of the inverted seven-spot well pattern and the five-spot one were 93.3% and 68.0%, respectively.

In general, compared to the five-spot well pattern, the swept area of the SP system in the inverted seven-spot well pattern was larger, but the intensity was weaker. When converting well pattern types, injection and production wells should be reasonably arranged so that the remaining oil and residual oil those are not recovered in the five-spot well pattern can be effectively recovered by utilizing the characteristics of the inverted seven-spot well pattern.

E. Main controlling factors and compatibility of SP flooding development in inverted seven-spot well pattern

After clarifying the differences in the migration law and the EOR ability of the SP system in the two types of well patterns, it is necessary to explore the main controlling factors of its development effect in the inverted seven-spot well pattern. By specifying the physical characteristics of crude oil, four crude oil systems with various viscosities were created, and 50 years of water flooding development were used to determine when different water cuts and oil recovery would occur, as shown in Table VII. It can be found that with the increase in crude oil viscosity, the water flooding recovery rate decreases obviously, and the

ultimate oil recovery decreased from 40.49% to 30.32%, with a decrease in about ten percentage points.

After gathering the fundamental information regarding water flooding, the timing for the subsequent development of the SP system was determined by reading the dynamic characteristic curve of water flooding development to determine the precise time of water flooding to each water cut under various crude oil viscosity conditions. In Fig. 13, the acquisition method was displayed.

The orthogonal experimental scheme of five factors and four levels was designed, as shown in Table VIII. By changing the fluid and injection parameters, the influence of different factors on the development effect of SP flooding was investigated. Among them, the regulation of the injection volume of the SP system is achieved by setting the subsequent water flooding timing according to the initial injection timing of the SP system plus the number of days required for the SP flooding. According to the production dynamic curves of 16 groups of experiments, the ultimate oil recovery after 50 years of development was obtained, as illustrated in Table VIII.

Combining four sets of water flooding primary data with 16 sets of SP flooding data, the fitting relationship between five factors and oil recovery was investigated. Four sets of water flooding also needed to define the SP parameters. The definition was as follows: the KYPAM concentration and the KPS concentration were defined as 0, the SP system injection timing was defined as 100% water cut, and the

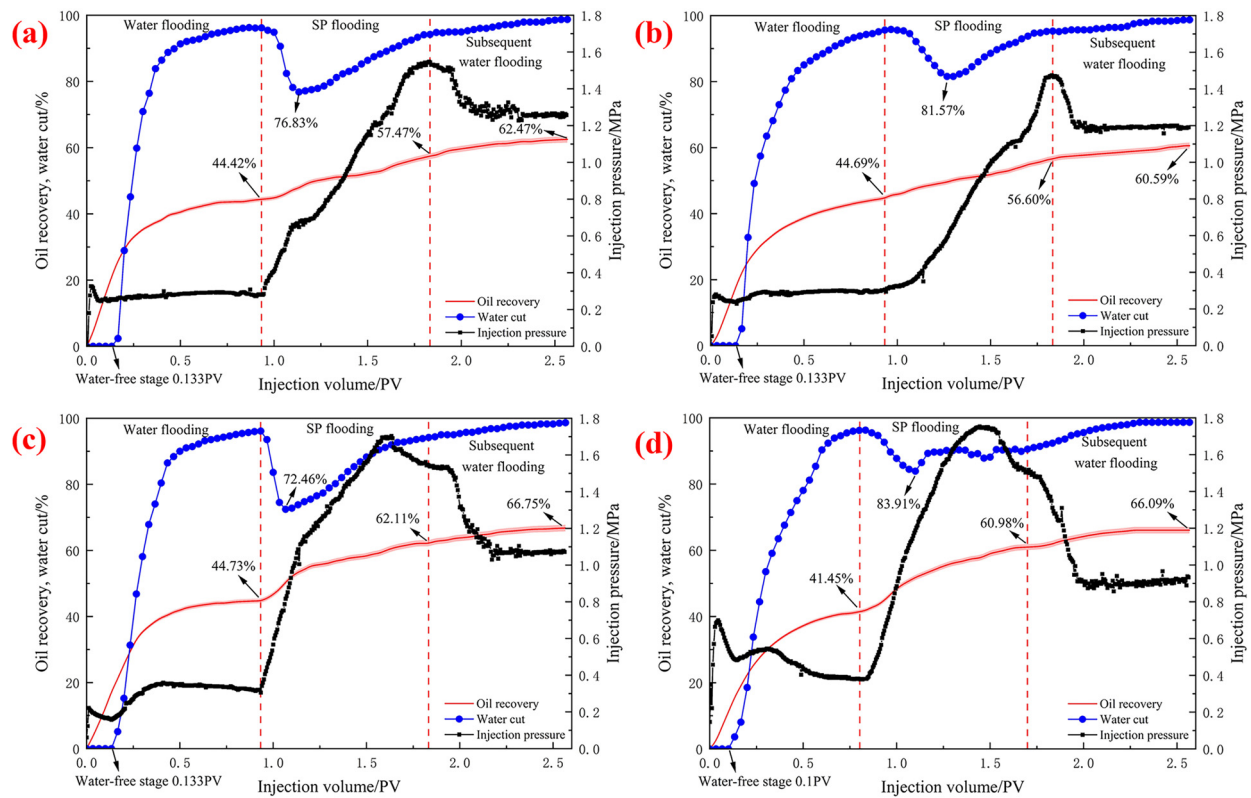


FIG. 10. Dynamic production curves under different SP system injection methods in the inverted seven-spot well pattern: (a) constant viscosity injection, (b) step-up viscosity injection, and (c) step-down viscosity injection; (d) dynamic production curves under step-down viscosity injection in the five-spot well pattern with equal control reserves.

injection amount was defined as 0. All 20 sets of data were employed to obtain the corresponding fitting formula by multiple linear regression. The outcome of the derived formula fitting is as follows.

$$Y = -0.041\,08X_1 + 28.456\,65X_2 + 9.938\,85X_3 + 0.015\,71X_4 - 1.038\,88X_5 + 36.887\,03. \quad (1)$$

In the formula, Y is the oil recovery, %; X_1 is the crude oil viscosity, mPa s; X_2 is the KYPAM concentration, wt. %; X_3 is the KPS concentration, wt. %; X_4 is the injection timing (water cut), %; X_5 is the injection volume, PV. The goodness of fit of the formula is $R^2 = 0.940\,38$.

The model prediction value and the formula fitting value are plotted in Fig. 14. It can be found that the error between the two is tiny, indicating that the formula has the excellent fitting ability at the significant level of 0.05.

Similarly, the correlation coefficient discrimination was performed using the previous 20 sets of data. The definition of the SP parameters of water flooding was the same as above. The Pearson correlation coefficient between each factor and crude oil recovery under a 95% confidence interval was obtained by a two-tailed significance test. As shown in Fig. 15, the orientation and the shape of the ellipse represent the positive or negative and the size of the correlation between two factors, respectively. The color depth and size of ellipses and numerical values can also represent the correlation. The cross in the

figure is an insignificant mark at the significant level of 0.05. The results of Pearson correlation coefficient showed that the correlation coefficients between oil recovery and injection volume, injection timing, surfactant concentration, polymer concentration, and crude oil viscosity were 0.54, -0.33 , 0.58, 0.70, and -0.61 , respectively. It demonstrates a positive correlation between oil recovery and injection volume, surfactant content, and polymer concentration. Injection timing and crude oil viscosity are negatively correlated with oil recovery. The polymer concentration is the most effective factor of SP flooding development in an inverted seven-spot well pattern, followed by crude oil viscosity, and surfactant concentration, while injection timing has the least influence on oil recovery.

The above results show that the injection volume and timing of the SP system have a relatively small impact on the ultimate oil recovery. Therefore, the influence of these two factors is not considered in the obtained fitting formula, that is, the X_4 and X_5 in the formula are fixed. Only the effects of crude oil viscosity, polymer concentration, and surfactant concentration on the enhanced oil recovery (ΔY) are studied:

$$\Delta Y = -0.041\,08X_1 + 28.456\,65X_2 + 9.938\,85X_3. \quad (2)$$

First, the oil viscosity X_1 was fixed, and then the polymer concentration X_2 and surfactant concentration X_3 were changed, respectively, to obtain the enhanced oil recovery ΔY under different conditions. According to different ΔY ranges, the adaptability of the SP system

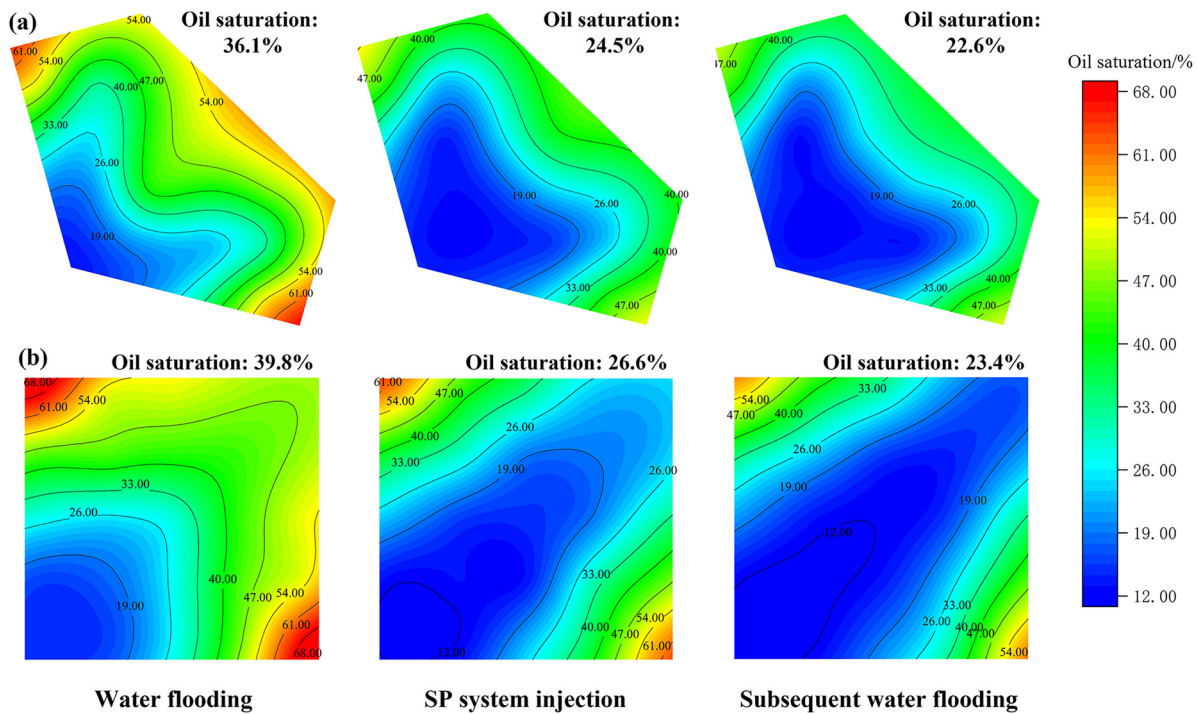


FIG. 11. Oil saturation distributions of (a) inverted seven-spot and (b) five-spot well patterns in different stages under the optimal SP system injection method.

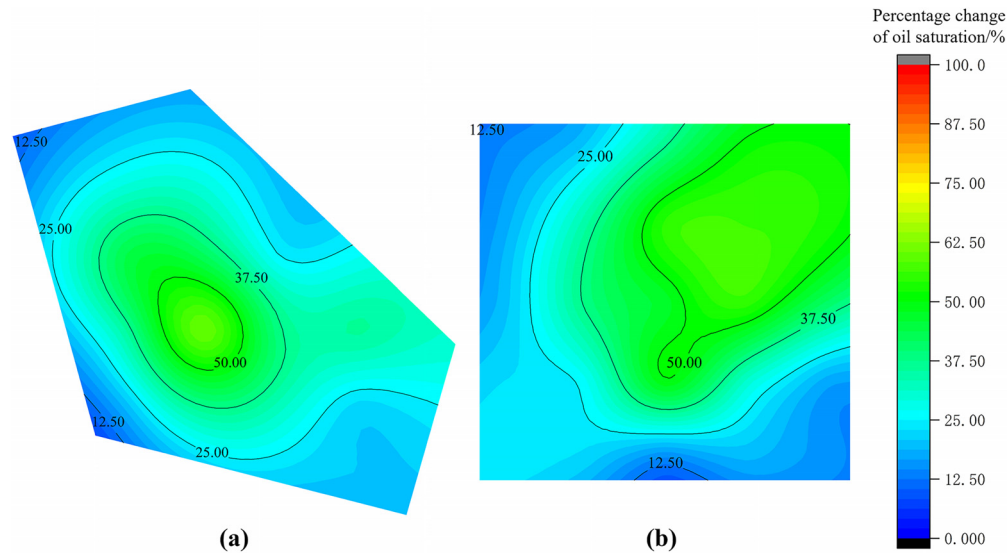


FIG. 12. Percentage change of oil saturation in (a) inverted seven-spot and (b) five-spot well patterns under optimal injection method.

was given. Taking the oil viscosity of 7.6 mPa s, KYPAM concentration of 0.5 wt. %, and KPS concentration of 0.3 wt. % as the example, the corresponding ΔY was calculated. That is, $\Delta Y = -0.041\,08 \times 7.6 + 28.456\,65 \times 0.5 + 9.938\,85 \times 0.3 = 16.897\,772$. This shows that the recovery rate under this condition is about 16.90%, which has an

excellent effect. Using the same method, the adaptability chart of SP flooding under different crude oil viscosity is established according to the EOR values, as shown in Fig. 16.

The adaptability of SP system to crude oil with different viscosity is significantly different. For crude oil with low viscosity, the SP system

TABLE VII. Oil recovery of each stage for water flooding development.

Oil viscosity/mPa s	Development time				
	Water cut 85%	Water cut 90%	Water cut 95%	Water cut 98%	50 years
7.6	30.95%	32.67%	35.23%	37.76%	40.49%
48	23.27%	25.04%	28.09%	32.83%	35.75%
100	20.84%	22.36%	25.26%	30.28%	33.07%
200	18.86%	20.09%	22.61%	27.82%	30.32%

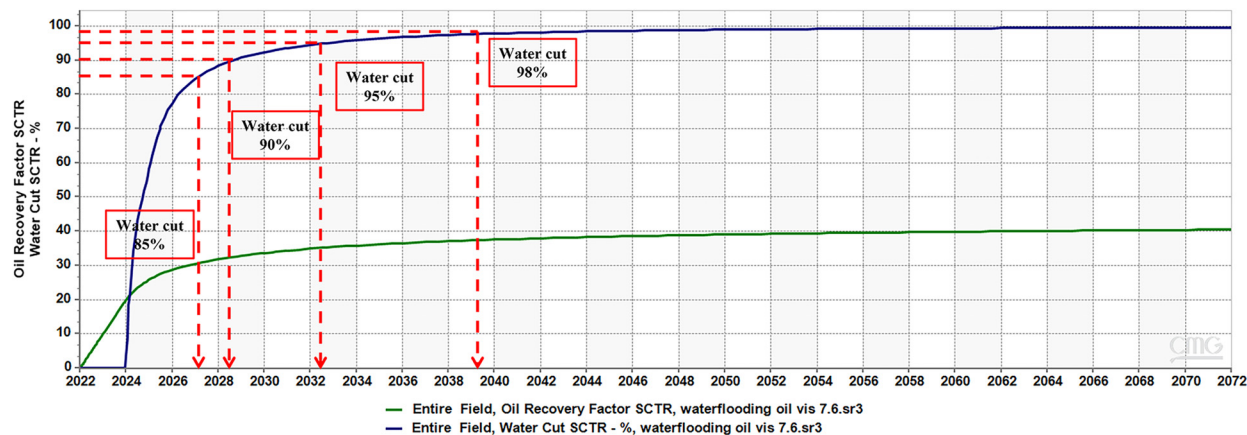


FIG. 13. Dynamic characteristic curve of water flooding development (oil viscosity 7.6 mPa s).

TABLE VIII. Ultimate oil recovery under the condition of five factors and four levels.

No.	Oil viscosity/mPa s	KYPAM concentration/wt. %	KPS concentration/wt. %	Injection timing/%	Injection volume/PV	Oil recovery/%
1	7.6	0.1	0.1	85	0.8	41.57
2	7.6	0.15	0.2	90	0.9	43.54
3	7.6	0.22	0.3	95	1	48.61
4	7.6	0.3	0.5	98	1.2	51.27
5	48	0.1	0.2	95	1.2	39.16
6	48	0.15	0.1	98	1	39.36
7	48	0.22	0.5	85	0.9	45.74
8	48	0.3	0.3	90	0.8	45.96
9	100	0.1	0.3	98	0.9	38.22
10	100	0.15	0.5	95	0.8	41.55
11	100	0.22	0.1	90	1.2	39.17
12	100	0.3	0.2	85	1	42.82
13	200	0.1	0.5	90	1	36.89
14	200	0.15	0.3	85	1.2	38.01
15	200	0.22	0.2	98	0.8	38.87
16	200	0.3	0.1	95	0.9	38.91

can displace crude oil more efficiently when KYPAM concentration is greater than 0.3 wt. %. With the gradual increase in oil viscosity, the oil recovery ability of the SP system gradually deteriorates. What is worse, when the concentration of the SP system is too low, high-viscosity crude oil cannot be effectively recovered (invalid injection). Therefore, in the actual operation process, the parameters of the SP system should be reasonably arranged according to the characteristics of crude oil.

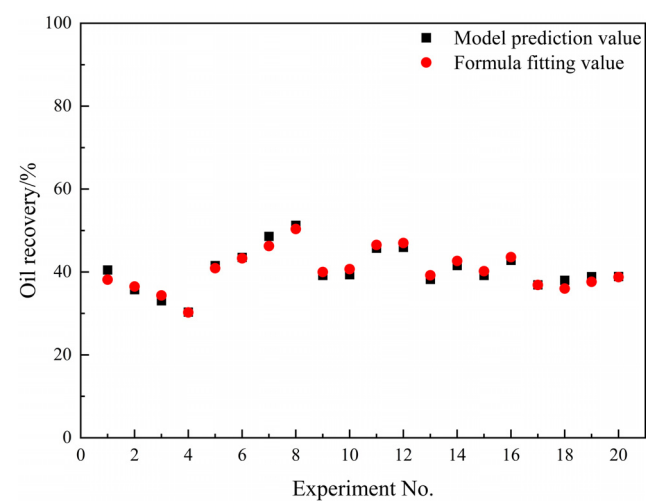


FIG. 14. Comparison of the predicted oil recovery with the formula fitting oil recovery.

IV. CONCLUSIONS

In this research, numerical simulation was used to determine the distributions of distinct sweep intensity regions in two well patterns. EOR abilities of the SP system under different sweep intensities were obtained by one-dimensional core oil displacement experiments. Tailor-made flat cores were used to simulate the inverted seven-spot well pattern. The sweeping capacity and the oil displacement efficiency under different SP system injection strategies were compared, and the adaptability of the SP system in two well patterns was evaluated. By establishing the numerical model of the inverted seven-spot well

pattern and using multiple linear regression and Pearson correlation coefficient, the recovery calculation formula under the influence of multiple factors was obtained, and the main controlling factors of the SP flooding development in the inverted seven-spot well pattern were clarified. The specific conclusions are as follows:

- (1) Compared with the five-spot well pattern, the overall pressure gradient of the inverted seven-spot well pattern is significantly lower, indicating a low-intensity swept state. The weak swept region accounts for 61%.
- (2) With the increase in sweep intensity, the oil-producing capacity of the SP system increases by about 15%–20%. Increasing the reservoir permeability and viscosity of the SP system can improve the oil displacement efficiency in each swept region. As far as the whole well pattern is concerned, the potential of the SP system in the inverted seven-spot well pattern is slightly lower, with a gap of roughly 4% points.
- (3) To further enhance the utilization degree of the well pattern, the viscosity and slug size of the SP system should be sufficiently increased within an acceptable range. The step-down viscosity injection strategy can achieve the highest swept area and EOR rate in the inverted seven-spot well pattern. The recovery value rises by 3.97% points compared to conventional constant viscosity injection.
- (4) The effective swept area of the SP system in the seven-spot well pattern (93.3%) is larger than that in the five-spot one (68.0%). The SP system in the seven-spot well pattern has better oil recovery capacity in the non-primary streamline area. However, the capacity in the primary streamline area is inferior to that in the five-spot well pattern. Although the SP system has a slightly lower EOR potential in the inverted seven-spot well pattern, there is minimal variation between the two well types' ultimate oil recovery in terms of the entire development cycle.

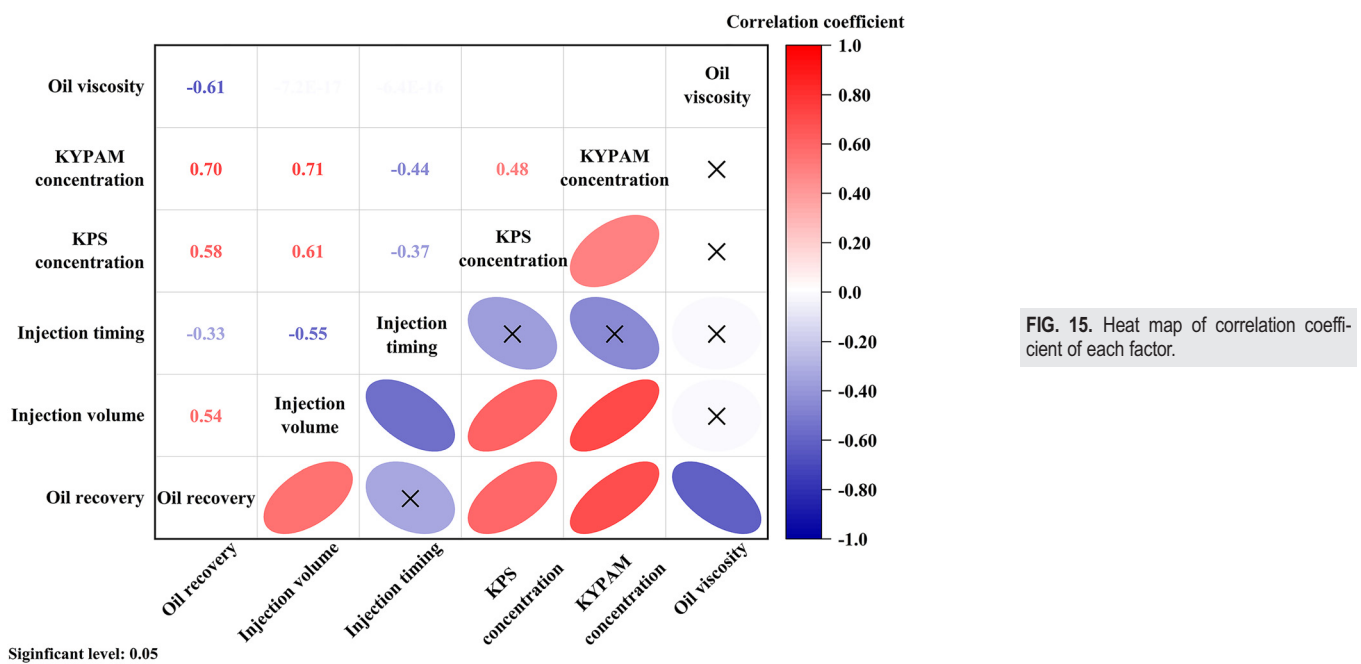


FIG. 15. Heat map of correlation coefficient of each factor.

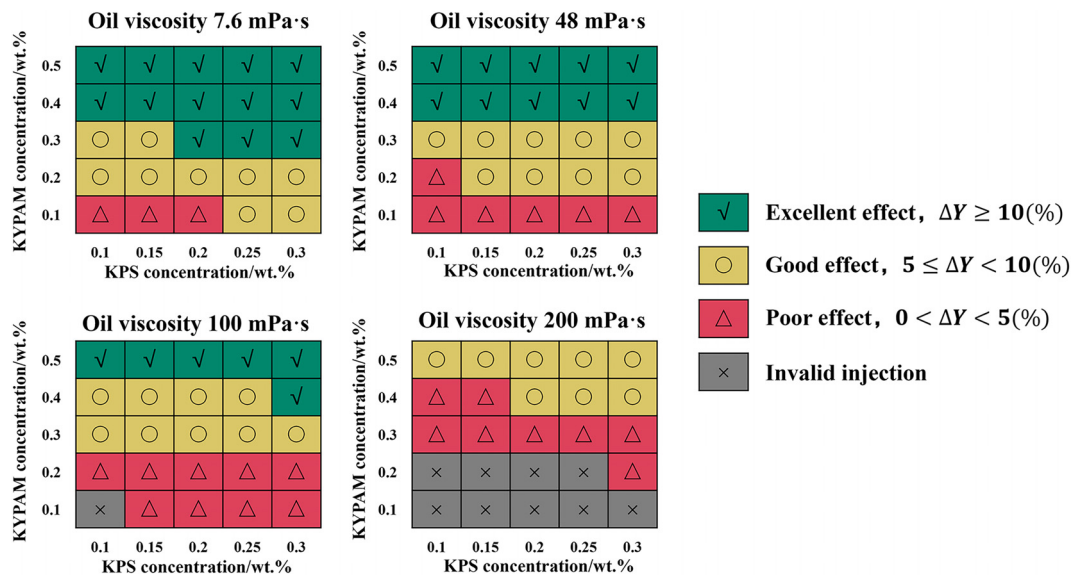


FIG. 16. Adaptability charts of SP system under different crude oil viscosity.

- (5) The oil recovery under the condition of five factors and four levels was obtained by the numerical model of the inverted seven-spot well pattern. The fitting formula between each factor and oil recovery was obtained: $Y = -0.04108X_1 + 28.45665X_2 + 9.93885X_3 + 0.01571X_4 - 1.03888X_5 + 36.88703$.
- (6) By a two-tailed significance test, the Pearson correlation coefficient between each factor and crude oil recovery within a 95% confidence interval was obtained. It was found that the main controlling factors for SP flooding development in the inverted seven-spot well pattern were, in order, polymer concentration, crude oil viscosity, and surfactant concentration. According to the increase in oil recovery, four different grades of EOR effects (excellent effect, good effect, poor effect, and invalid injection) are divided, and the compatibility chart is established.

The results may serve as a guide for the popularization and deployment of SP flooding technology in the inverted seven-spot well pattern, in addition to offering technical assistance for the design of injection parameters for SP flooding in Karamay oilfields.

SUPPLEMENTARY MATERIAL

See the supplementary material related to this article in the online version.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Xuechen Tang: Conceptualization (lead); Data curation (lead); Formal analysis (lead); Investigation (lead); Methodology (lead); Software (equal); Visualization (lead); Writing – original draft (lead); Writing – review & editing (lead). **Yiqiang Li:** Funding acquisition (lead); Project administration (equal); Resources (lead); Supervision (lead); Validation (equal). **Jinxin Cao:** Formal analysis (supporting); Software (equal). **Zheyu Liu:** Conceptualization (equal); Project administration (equal); Validation (equal). **Xin Chen:** Conceptualization (supporting); Methodology (equal). **Li Liu:** Data curation (supporting); Investigation (supporting). **Yaqian Zhang:** Investigation (supporting). **Qihang Li:** Data curation (supporting).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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